Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Owned scheduling, **fault-tolerant placement**, and orchestration for compute infrastructure running **VMs in large-scale clusters** across AWS, GCP, and CoreWeave.
- Kept fleet utilization up through **node loss** and provider disruptions by moving workloads off unhealthy capacity onto healthy nodes, so **long-running jobs** kept running.
- Built fleet **observability** and **autoscaling** with **Prometheus/Grafana**, and ran distributed workloads on **Slurm** and **Kubernetes** over **InfiniBand/RoCE** networking.
- Built a **distributed executor** exposing typed **REST APIs** with an **S3** artifact store and checkpointing, so interrupted workflows **resume from the last committed step** at sub-200ms p99.

QureAI | *AI Engineer* Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by moving synchronous services to **async processing** with autoscaling.
- Built **CI/CD** with automated quality gates, failure alerting, and **automated rollback** for safe rollouts.
- Built an **SRE incident agent** that triages production alerts and escalates data questions to a SQL agent over **A2A**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Raised service availability to **99.9%** and cut **p99 latency 32%** by building **Python/C++** serving infrastructure with health checks, retry policies, and **automated failure recovery**.
- Raised throughput **2.4x** and cut **GPU memory 27%** by **profiling** compute and memory bottlenecks and optimizing batching, request scheduling, and concurrency in **C++** and Python services.
- Cut production escalations **60%** in one quarter by standardizing retry, timeout, and failure contracts across **25+ distributed services**, driving **root cause analysis**.
- Led **design and code reviews** for services serving **1M+ users** at **99.9% uptime**, and mentored engineers on reliability practices.

Projects

Distributed NoSQL Database `github.com/ganeshhgupta/distributed-kv-store`

- **Fault-tolerant** distributed database that confirms writes only after a **majority of nodes** store them, recovers crashed nodes from a **write-ahead log**, and resyncs partitioned replicas via **leader-follower replication**.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Observability platform that traces model calls, latency, and token usage, with a **FastAPI** backend, **PostgreSQL**, and a **React/TypeScript** dashboard.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: C++, Python, Java, SQL

Distributed Systems: Scheduling, Fault-Tolerant Placement, Replication, Write-Ahead Logging, Concurrency, Microservices

Infrastructure: GCP, AWS, Kubernetes, Slurm, Docker, InfiniBand/RoCE, NVIDIA GPUs, CI/CD

Reliability: Prometheus, Grafana, Profiling, Debugging, Incident Response, Root Cause Analysis, DS&A

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer